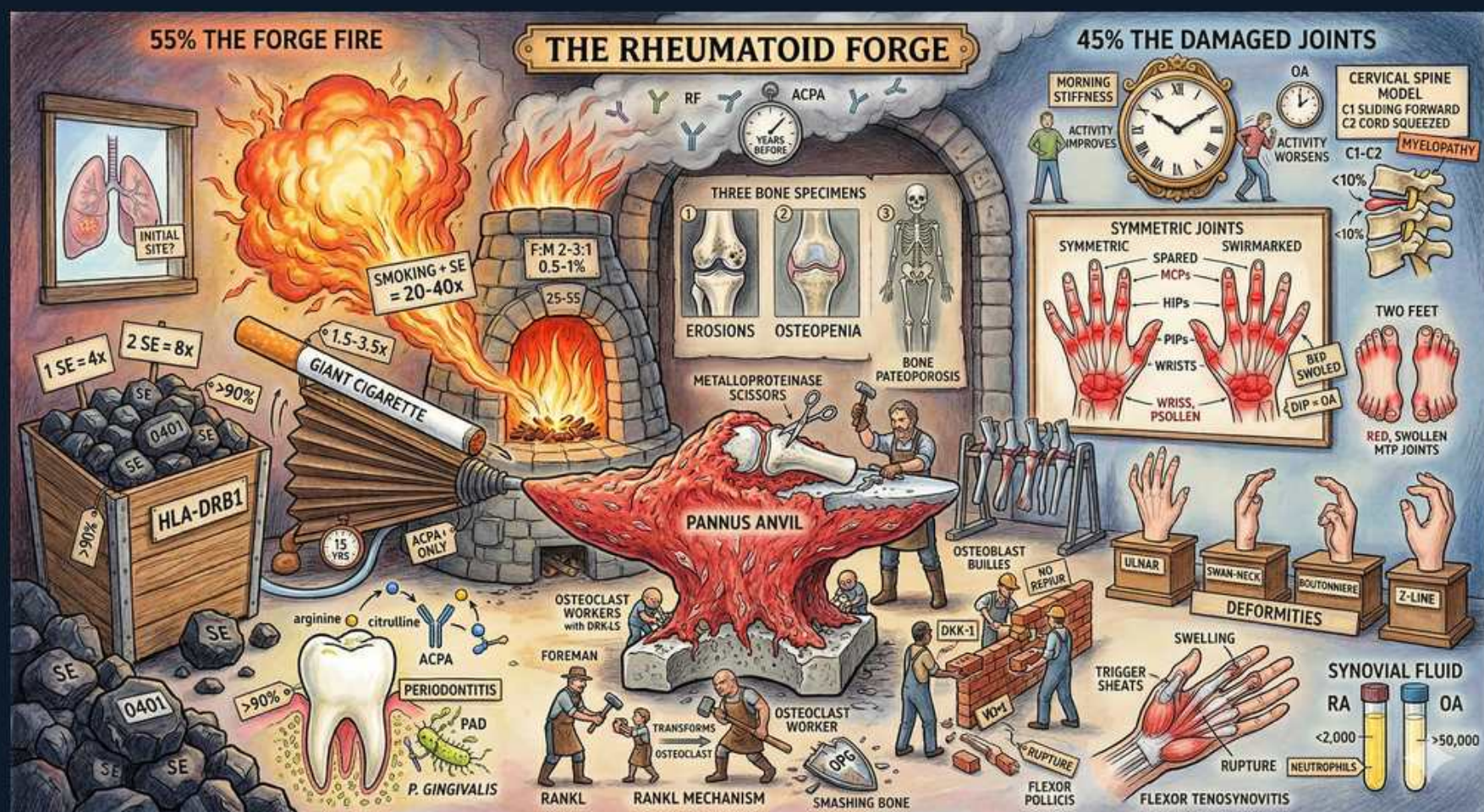


Rheumatoid Arthritis — Pathogenesis & Articular Disease



1. HLA-DRB1 / shared epitope

WHAT YOU SEE

Coal blocks labeled HLA-DRB1 *0401 / *0404 feeding the forge

WHAT IT ENCODES

The strongest GENETIC risk for RA is the HLA-DRB1 'shared epitope' (a conserved amino-acid sequence in alleles such as *0401 and *0404). This class II MHC motif preferentially presents citrullinated peptides to CD4+ T cells, driving anti-CCP production — it is most strongly linked to seropositive (anti-CCP+) RA. Shared epitope plus smoking is a powerful gene–environment interaction multiplying risk.

BOARD TRAP

A board question swaps in HLA-B27 as the answer for RA's genetic association. The discriminator: HLA-B27 belongs to the SERONEGATIVE spondyloarthropathies (ankylosing spondylitis, reactive/psoriatic arthritis, IBD-associated); RA's association is HLA-DRB1 shared epitope, an MHC class II (not class I) allele.

2. Smoking

WHAT YOU SEE

Giant cigarette feeding the forge fire, RR up to 20-40x with shared epitope

WHAT IT ENCODES

Cigarette smoking is the major environmental trigger for RA: it induces peptidylarginine deiminase (PAD) in the lung, citrullinating proteins and generating anti-CCP/ACPA in genetically susceptible (shared-epitope) hosts. The gene–environment synergy is striking — combined shared epitope + smoking raises risk of seropositive RA up to ~20–40 fold over neither factor.

BOARD TRAP

Treating smoking as merely a generic risk factor. The discriminator: smoking is a MODIFIABLE driver of citrullination/anti-CCP AND it blunts response to anti-TNF (and other) therapy — so smoking cessation is part of RA management, and continued smoking predicts worse treatment response and severity.

3. Citrullination / anti-CCP

WHAT YOU SEE

Arginine to citrulline conversion making ACPA antibodies

WHAT IT ENCODES

Peptidylarginine deiminase (PAD) enzymes convert arginine residues to CITRULLINE (post-translational modification). These citrullinated self-proteins become neoantigens, and antibodies against them (ACPA, detected clinically as anti-CCP) are highly specific for RA. ACPA can be present years before symptom onset and predicts erosive, aggressive disease.

BOARD TRAP

Confusing the test characteristics: the wrong answer calls anti-CCP the most SENSITIVE test or claims it is less specific than RF. The discriminator: anti-CCP is the most SPECIFIC marker (~95%), whereas RF is more sensitive but far less specific (also positive in HCV, Sjögren, endocarditis). Specificity ≠ sensitivity.

4. P. gingivalis / periodontitis

WHAT YOU SEE

Gum disease bacteria at the base generating citrullination

WHAT IT ENCODES

Porphyromonas gingivalis, the periodontitis pathogen, uniquely expresses its own bacterial PAD enzyme that citrullinates host proteins at the mucosal (gingival) surface — implicating periodontitis as a MUCOSAL trigger that seeds anti-CCP responses before joint disease. This parallels the lung (smoking) as a site where citrullination initiates RA autoimmunity in susceptible hosts.

BOARD TRAP

Dismissing periodontal disease as unrelated, or picking an unrelated organism. The discriminator: P. GINGIVALIS is the specific periodontal organism mechanistically linked to RA via bacterial citrullination; the broader teaching point is that RA autoimmunity often begins at MUCOSAL surfaces (gums, lungs), not in the joint itself.

5. Pannus

WHAT YOU SEE

Red inflamed tissue mass labeled PANNUS ANVIL at center

WHAT IT ENCODES

Pannus is the hallmark lesion of RA: hyperplastic, inflamed synovium transformed into an invasive, tumor-like tissue of activated fibroblast-like synoviocytes, macrophages, T/B cells, and new blood vessels. It physically invades and ERODES the adjacent articular cartilage and subchondral bone at the joint margins, secreting cytokines (TNF, IL-6, IL-1) and proteases that destroy the joint.

BOARD TRAP

A stem describes joint destruction and the wrong answer attributes it to primary cartilage wear (OA) or crystal deposition. The discriminator: in RA the destruction is driven by an INVASIVE inflammatory pannus (immune-mediated synovial proliferation), not the 'wear-and-tear' cartilage loss of osteoarthritis or crystal-induced damage of gout.

6. RANKL / osteoclasts

WHAT YOU SEE

Osteoclast workers smashing bone via RANKL mechanism

WHAT IT ENCODES

Inflammatory cytokines (TNF, IL-6, IL-17) in the RA synovium upregulate RANKL on synovial fibroblasts and T cells. RANKL binds RANK on osteoclast precursors, driving their differentiation and activation — the final common pathway for the periarticular bony EROSIONS and juxta-articular osteopenia of RA. (Denosumab, an anti-RANKL antibody, can reduce erosion progression.)

BOARD TRAP

Attributing RA bone erosions to direct osteoblast failure or to systemic osteoporosis alone. The discriminator: the erosions are OSTEOCLAST-mediated via the RANKL/RANK axis driven by synovial inflammation — so controlling inflammation (and RANKL) protects bone, whereas misframing it as a primary bone-formation defect is incorrect.

7. Metalloproteinase (MMP)

WHAT YOU SEE

Metalloproteinase scissors cutting between bone specimens

WHAT IT ENCODES

Matrix metalloproteinases (MMPs), secreted by activated synoviocytes and chondrocytes in response to TNF/IL-1/IL-6, enzymatically degrade the collagen and proteoglycan matrix of articular CARTILAGE. Together with osteoclast-mediated bony erosion, MMP-driven cartilage breakdown produces the joint-space narrowing and destruction seen radiographically.

BOARD TRAP

Picturing two separate diseases for cartilage vs bone. The discriminator: RA joint destruction has TWO complementary arms driven by the same cytokine milieu — MMPs destroy CARTILAGE while RANKL/osteoclasts erode BONE; both are downstream of TNF/IL-6, which is why cytokine-targeted biologics halt structural damage.

8. Erosions / osteopenia

WHAT YOU SEE

Three bone specimens showing erosions and periarticular osteopenia

WHAT IT ENCODES

The hallmark radiographic findings of RA are MARGINAL (juxta-articular) bony erosions and PERIARTICULAR osteopenia, with symmetric joint-space narrowing — classically at the MCPs, PIPs, wrists, and MTPs. Erosions occur at the 'bare areas' where synovium contacts bone unprotected by cartilage. Early films may be normal; MRI/ultrasound detect erosions sooner.

BOARD TRAP

Choosing OA-type radiographic findings for RA. The discriminator: RA shows PERIARTICULAR osteopenia, marginal erosions, and symmetric joint-space narrowing — NOT the subchondral SCLEROSIS, OSTEOPHYTES, and asymmetric narrowing of osteoarthritis. Mixing up these radiographic signatures is the classic trap.

9. Symmetric small joints (MCP/PIP)

WHAT YOU SEE

Red symmetric hands showing MCP, PIP, wrists involved, DIP spared

WHAT IT ENCODES

RA is a SYMMETRIC polyarthritis of the SMALL joints — characteristically the MCPs, PIPs, and wrists of both hands — that classically SPARES the distal interphalangeal (DIP) joints. Involvement of ≥3 joint areas, small-joint predominance, and symmetry contribute to the 2010 ACR/EULAR classification criteria along with serology, acute-phase reactants, and symptom duration ≥6 weeks.

BOARD TRAP

A patient with prominent DIP involvement and the wrong answer is RA. The discriminator: RA SPARES the DIP joints — DIP arthritis points instead to OSTEOARTHRITIS (Heberden nodes) or PSORIATIC arthritis (with nail pitting/dactylitis). Symmetric MCP/PIP/wrist disease sparing DIPs is the RA pattern.

10. Morning stiffness >1h

WHAT YOU SEE

Clock at top showing morning stiffness worsening with activity rest

WHAT IT ENCODES

Morning stiffness lasting MORE than 1 hour that IMPROVES with activity/use is the cardinal INFLAMMATORY symptom of RA (gel phenomenon). It reflects active synovitis and contrasts sharply with mechanical joint disease. Duration and presence of stiffness help separate inflammatory from non-inflammatory arthropathy in the history.

BOARD TRAP

A vignette gives brief (<30 min) stiffness that WORSENS with use, and the wrong answer is RA. The discriminator: inflammatory (RA) stiffness lasts >1 hour and gets BETTER with activity; osteoarthritis stiffness is short (<30 min) and gets WORSE with use as the day goes on. The pattern of stiffness, not just its presence, is the key.

11. Cervical spine / atlantoaxial

WHAT YOU SEE

Cervical spine model C1 sliding forward, C2 cord squeezed

WHAT IT ENCODES

RA synovitis of the C1–C2 (atlantoaxial) joint and erosion of the transverse ligament/odontoid can cause ATLANTOAXIAL SUBLUXATION, risking spinal cord compression. Warning signs: neck pain, occipital headache, and myelopathy (limb weakness, paresthesias, hyperreflexia, gait/bladder changes). Flexion–extension cervical films (or MRI) are obtained, and this must be assessed BEFORE intubation/neck manipulation to avoid catastrophic cord injury.

BOARD TRAP

Forgetting the cervical spine in pre-op clearance of a long-standing RA patient. The discriminator: RA SPARES the thoracolumbar spine but threatens the CERVICAL (C1–C2) spine — failing to obtain flexion–extension views/MRI before intubation risks cord compression during neck extension. The dangerous level is atlantoaxial, not the lumbar spine.

12. MTP joints / two feet

WHAT YOU SEE

Two feet with red swollen MTP joints

WHAT IT ENCODES

The metatarsophalangeal (MTP) joints of the feet are commonly and EARLY involved in RA — often among the first joints affected. A positive 'squeeze test' (pain on transverse compression across the MTP or MCP heads) is a sensitive early bedside sign of inflammatory synovitis. Forefoot pain/'walking on pebbles' may precede hand complaints.

BOARD TRAP

Overlooking the feet because the classic image is the hands. The discriminator: MTP involvement is frequently an EARLY sign of RA (positive MTP squeeze test), so dismissing forefoot symptoms as plantar/mechanical and waiting for hand findings can delay diagnosis.

13. Deformities (swan-neck, boutonniere, ulnar)

WHAT YOU SEE

Hands showing trigger sheaths, ulnar deviation, deformities

WHAT IT ENCODES

LATE hand deformities from chronic tendon/ligament and joint damage include ULNAR deviation of the fingers at the MCPs, SWAN-NECK deformity (PIP hyperextension + DIP flexion), BOUTONNIÈRE deformity (PIP flexion + DIP hyperextension), and Z-thumb. These reflect long-standing untreated disease — early effective DMARD therapy aims to prevent them.

BOARD TRAP

Mixing up swan-neck and boutonnière, or treating these deformities as early/diagnostic features. The discriminator: SWAN-NECK = PIP hyperextended, DIP flexed; BOUTONNIÈRE = PIP flexed, DIP hyperextended (the PIP 'buttonholes' through the extensor hood). They are LATE complications of inadequately controlled RA, not early diagnostic signs.

14. Synovial fluid (inflammatory)

WHAT YOU SEE

Vials comparing RA vs OA synovial fluid, WBC >50,000 marked

WHAT IT ENCODES

RA synovial fluid is INFLAMMATORY (Class II): cloudy, WBC roughly 2,000–50,000/μL with a neutrophil predominance, and crucially NO crystals and STERILE Gram stain/culture. Arthrocentesis is performed mainly to EXCLUDE septic arthritis and crystal disease, since RA itself has no pathognomonic fluid finding. Markedly elevated counts (>50,000, especially >100,000) should raise concern for infection.

BOARD TRAP

Reading an RA-range count as septic, or expecting RA fluid to be non-inflammatory. The discriminator: RA fluid is inflammatory (~2,000–50,000 WBC, neutrophilic, no crystals, sterile) — distinctly higher than the <2,000 non-inflammatory count of OSTEOARTHRITIS, but if WBC is >50,000 (especially >100,000) you must rule out SEPTIC arthritis even in a known RA patient.